AI ASSITED CODING LAB TEST-02

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BATCH-18

SET-L

1Q.

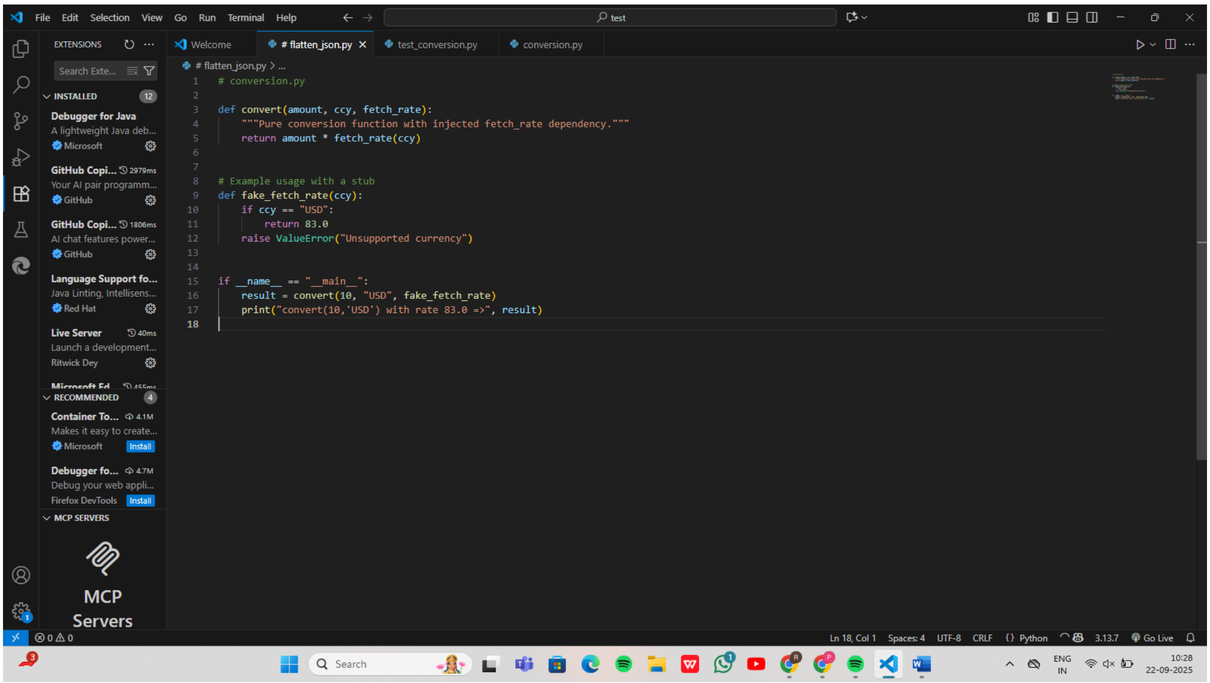
PROMPT :-

Write Python code for a pure function convert(amount, ccy, fetch_rate) that multiplies amount * fetch_rate(ccy) .fetch_rate should be injected as a dependency.Also, provide a stubbed fetch_rate function in the test so that

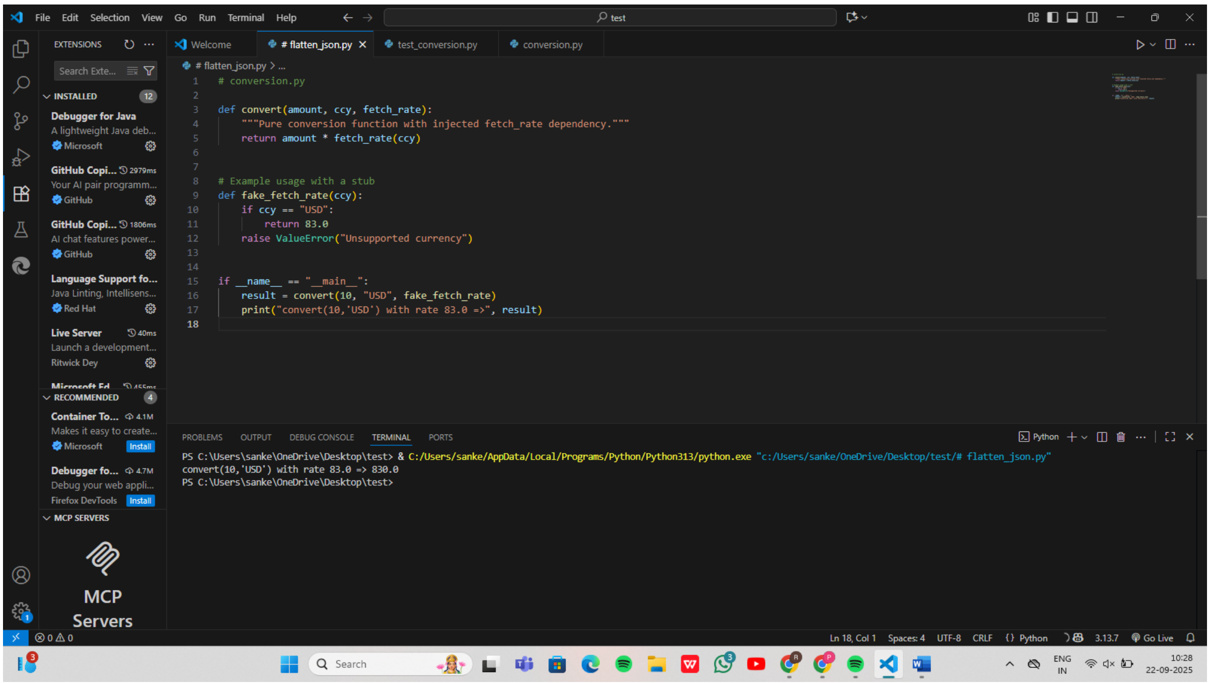
For USD, rate = 83.0

- convert(10, "USD", fake_fetch_rate) => 830.0

CODE



OUPTUT:-



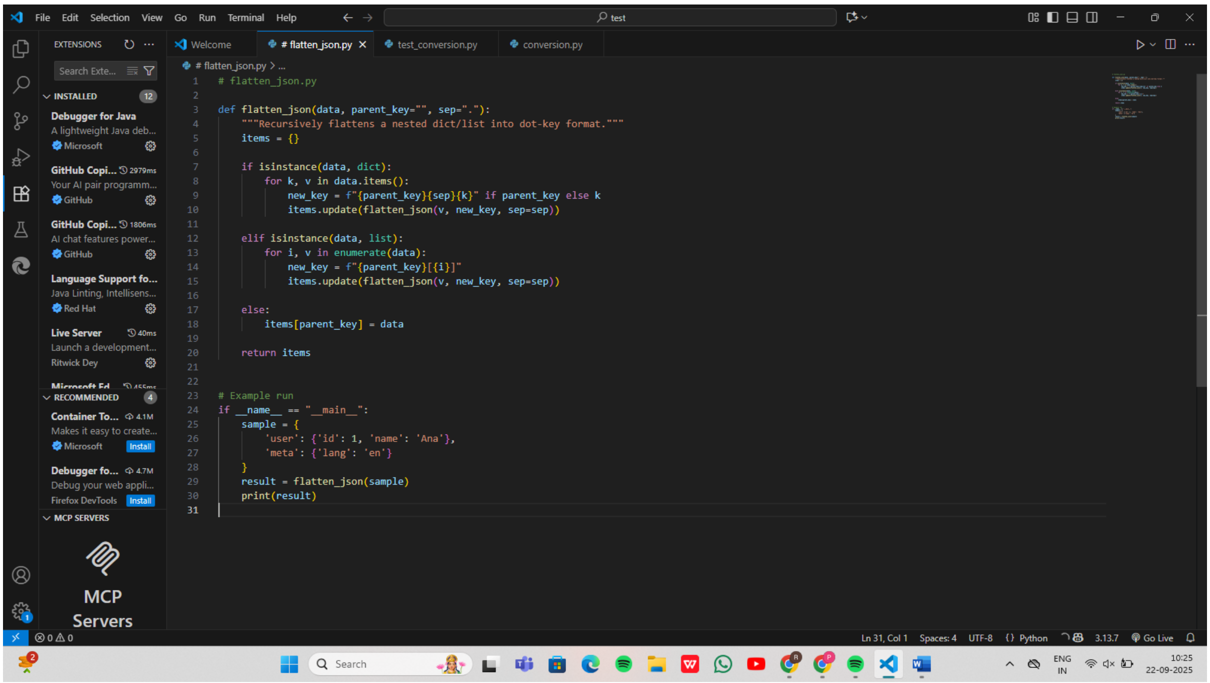
2Q.

PROMPT:-

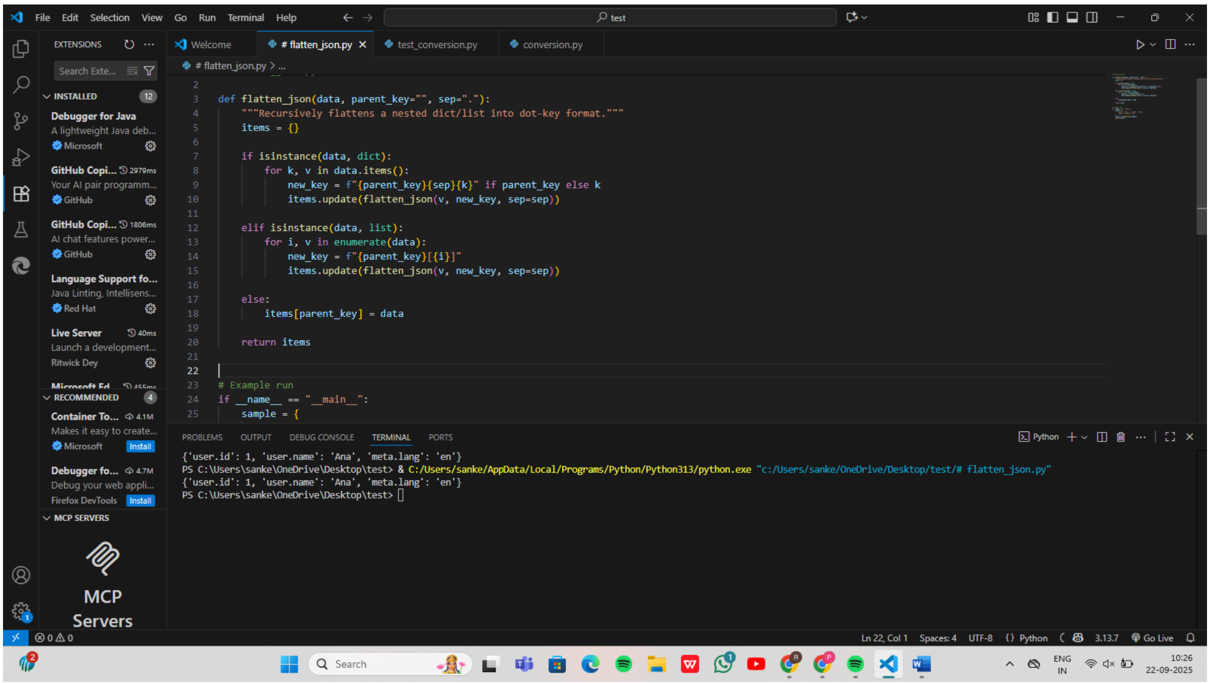
Write Python code for a function flatten json(data) that:

- Flattens nested dicts and lists into a flat dict
- Uses dot notation for dict keys
- Uses [index] for list elements
- Returns a flat dict

CODE:-



OUTPUT:-


File Edit Selection View Go Run Terminal Help
test
EXTENSIONS
INSTALLED
Debugger for Java
A lightweight Java deb...
Microsoft
GitHub Copi...
Your AI pair programm...
GitHub
GitHub Copi...
AI chat features power...
GitHub
Language Support fo...
Java Linting, Intellisens...
Red Hat
Live Server
Launch a development...
Ritwick Dey
RECOMMENDED
Container To...
Makes it easy to create...
Microsoft
Install
Debugger fo...
Debug your web appli...
Firefox DevTools
Install
MCP SERVERS
MCP Servers
Welcome
flatten_json.py
test_conversion.py
conversion.py
def flatten_json(data, parent_key="", sep="."):
"""Recursively flattens a nested dict/list into dot-key format."""
items = {}
if isinstance(data, dict):
for k, v in data.items():
new_key = f"{parent_key}{sep}{k}" if parent_key else k
items.update(flatten_json(v, new_key, sep=sep))
elif isinstance(data, list):
for i, v in enumerate(data):
new_key = f"{parent_key}[{i}]"
items.update(flatten_json(v, new_key, sep=sep))
else:
items[parent_key] = data
return items
# Example run
if __name__ == "__main__":
sample = {
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
{'user.id': 1, 'user.name': 'Ana', 'meta.lang': 'en'}
PS C:\Users\sanke\OneDrive\Desktop\test> & C:/Users/sanke/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/sanke/OneDrive/Desktop/test/# flatten_json.py"
{'user.id': 1, 'user.name': 'Ana', 'meta.lang': 'en'}
PS C:\Users\sanke\OneDrive\Desktop\test>
Ln 22, Col 1 Spaces: 4 UTF-8 Python 3.13.7 Go Live
Search
ENG IN
10:26
22-09-2025